

Spreading 36 Points in the Plane

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Abstract

We place 36 points in the plane so that the squared ratio of the largest to the smallest distance between them is at most 32.83385+. Erich Friedman’s table of best known configurations lists n up to 31. Deleting a point from our 37-point configuration gives 33.16344; this one is smaller. The ratio is computed exactly from the published coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 36 points in the plane with*

$$\left(\frac{\max d}{\min d}\right)^2 = 32.83385\dots$$

The exact value, a rational number, is given in Appendix A.

2 Context

Erich Friedman’s page *Minimizing the Ratio of Maximum to Minimum Distance* lists the best known configurations for $n \leq 31$. The value for $n = 36$ is not on the page. Deleting a point from our 37-point configuration gives 33.16344; this one is smaller.

3 Verification

The coordinates are decimals with 12 digits after the point, so they are exact rationals; all squared distances, and their largest-to-smallest ratio, are computed with integers. Run

```
python friedman-maxmin2d/verify.py friedman-maxmin2d/coords/n36.txt
```

in <https://github.com/Ge-limin/math>.

4 How it was found

Random starts of a constrained local optimizer (SLSQP), then basin hopping seeded from the neighbouring sizes (`friedman-maxmin2d/`).

A Exact value

```
numerator:
32833850400142367679663341
denominator:
999999999998385733200960
```

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

B Coordinates

36 points, x and y per line. File: `friedman-maxmin2d/coords/n36.txt`.

```
-9.076607860377 8.769205225320
-6.330093929933 8.012593040943
-7.132986014560 8.610849682434
-5.731784675492 5.127922356374
-7.053925263340 6.589115543741
-10.071554203608 8.869613263299
-7.398427721017 4.118345682216
-6.448371486720 4.430424440983
-7.714971782828 9.839678591344
-8.759332686516 6.286484545700
-8.704815861830 9.697521286701
-10.655987927999 8.058122473795
-6.155369841748 7.027975606094
-8.888040121382 5.294801937095
-10.712437421209 6.100592517049
-8.086763781374 8.911362529963
-6.266114112633 5.973198673683
-9.717486436837 6.000230478923
-5.959939457703 8.941563259350
-7.819266930836 5.945491327296
-7.006534497993 5.296862515098
-5.273339561921 6.093193221304
-7.228649351524 7.573732978590
-9.661036943627 7.957760435669
-7.947974365702 4.953808718691
-8.574531168249 4.174432960034
-5.157078704113 7.086411934852
-7.993991019020 6.930108762145
-10.887161509393 7.085209951898
-10.272174073807 5.168171811885
-5.331802792297 8.071029369701
-8.934056774701 7.271101980549
-9.892210525021 6.984847913771
-8.168715107205 7.914726196994
-6.760802703955 9.540410536042
-9.514596923930 4.515426178438
```